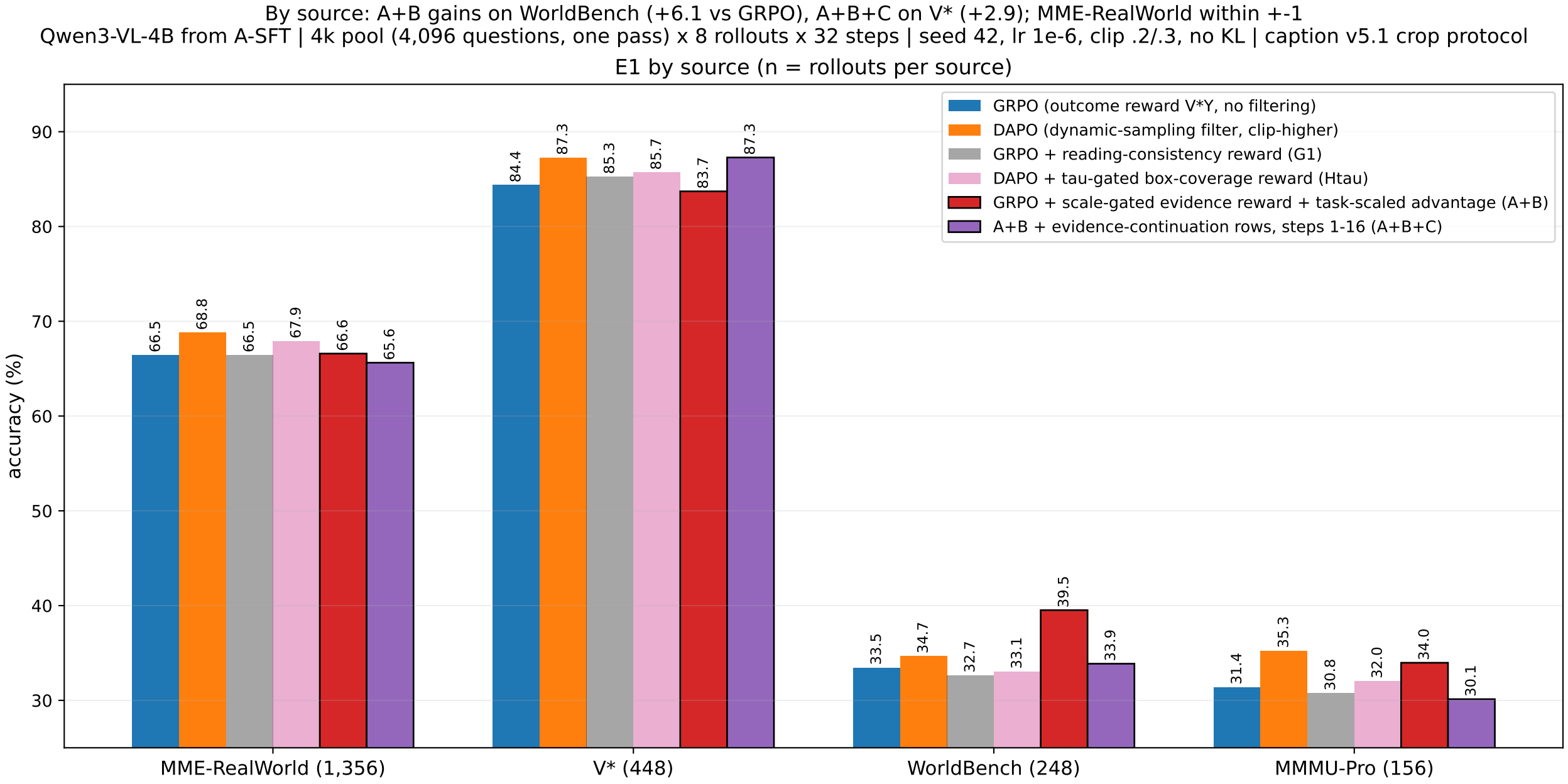
By source: A+B gains on WorldBench (+6.1 vs GRPO), A+B+C on V* (+2.9); MME-RealWorld within +-1
Qwen3-VL-4B from A-SFT | 4k pool (4,096 questions, one pass) x 8 rollouts x 32 steps | seed 42, lr 1e-6, clip .2/.3, no KL | caption v5.1 crop protocol
E1 by source (n = rollouts per source)
GRPO (outcome reward V*Y, no filtering)
DAPO (dynamic-sampling filter, clip-higher)
GRPO + reading-consistency reward (G1)
DAPO + tau-gated box-coverage reward (Htau)
GRPO + scale-gated evidence reward + task-scaled advantage (A+B)
A+B + evidence-continuation rows, steps 1-16 (A+B+C)
accuracy (%)
90
80
70
60
50
40
30
66.5
68.8
66.5
67.9
66.6
65.6
84.4
87.3
85.3
85.7
83.7
87.3
33.5
34.7
32.7
33.1
39.5
33.9
31.4
35.3
30.8
32.0
34.0
30.1
MME-RealWorld (1,356)
V* (448)
WorldBench (248)
MMMU-Pro (156)